MAL6150

Quiz 1: Questions and Brief Answers

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1. [2 marks] A fair coin is tossed repeatedly until a head appears for the first time. If a head never appears, the tossing continues forever.

(a) Write down the sample space Ω explicitly.

(b) Write down the event that a head appears for the first time on some even-numbered toss.

Brief answer.

$$\Omega = \{T^n H : n = 0, 1, 2, \ldots\} \cup \{T^\infty\}, \qquad T^\infty = TTT\cdots.$$

The event in (b) is

$$\{T^{2k-1}H : k \geq 1\} = \{TH, TTTH, TTTTTH, \ldots\}.$$

2. [2 marks] Let

$$\Omega = \{1, 2, 3, 4, 5\} \qquad \text{and} \qquad A = \{1, 2\}.$$

Find the smallest σ-algebra on Ω that contains A, and prove that it is indeed the smallest such σ-algebra.

Brief answer.

$$\mathcal{S} = \{\varnothing, \{1, 2\}, \{3, 4, 5\}, \Omega\}.$$

This is a σ-algebra. Any σ-algebra containing A must also contain $A^c = \{3, 4, 5\}$, $\varnothing$, and Ω, so it must contain $\mathcal{S}$.

3. [3 marks] Let Ω be a set, let $\mathcal{S}'$ be a σ-algebra on Ω', and let

$$f : \Omega \to \Omega'$$

be a function.

(a) For $A \subseteq \Omega'$, show that

$$\left(f^{-1}(A)\right)^c = f^{-1}(A^c).$$

(b) For sets $A_1, A_2, \ldots \subseteq \Omega'$, show that

$$\bigcup_{n=1}^{\infty} f^{-1}(A_n) = f^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right).$$

(c) Define

$$\mathcal{S} = \{f^{-1}(A) : A \in \mathcal{S}'\}.$$

Using parts (a) and (b), show that $\mathcal{S}$ is a σ-algebra on Ω.

Brief answer. For (a), $x \notin f^{-1}(A)$ iff $f(x) \notin A$, which is equivalent to $x \in f^{-1}(A^c)$. For (b), x belongs to the left-hand side iff $f(x) \in A_n$ for some n, which is equivalent to belonging to the right-hand side. Finally,

$$f^{-1}(\Omega') = \Omega,$$

and (a) and (b) show that $\mathcal{S}$ is closed under complements and countable unions. Hence $\mathcal{S}$ is a σ-algebra.

4. [1 mark] Let

$$\Omega = \{1, 2, 3, \dots\}$$

and suppose

$$P(\{n\}) = \frac{2}{3^n}, \qquad n = 1, 2, \dots.$$

(a) Verify that the probabilities of all one-point events add up to 1.

(b) Find the probability that n is even.

Brief answer.

$$\sum_{n=1}^{\infty} \frac{2}{3^n} = 1, \qquad P(n \text{ even}) = \sum_{k=1}^{\infty} \frac{2}{3^{2k}} = \frac{1}{4}.$$

5. [2 marks] A fair die is rolled once. Let

$$A = \{2, 4, 6\}, \qquad B = \{1, 2, 3, 4\}.$$

Are A and B independent? Justify your answer using the definition of independence.

Brief answer.

$$P(A) = \frac{1}{2}, \qquad P(B) = \frac{2}{3}, \qquad P(A \cap B) = P(\{2, 4\}) = \frac{1}{3}.$$

Since $\frac{1}{3} = \frac{1}{2} \cdot \frac{2}{3}$, the events are independent.

Bonus question. [2 marks] Let $\mathcal{S}_1$ and $\mathcal{S}_2$ be two σ-algebras on the same set Ω. Show that

$$\mathcal{S}_1 \cup \mathcal{S}_2$$

is a σ-algebra on Ω if and only if

$$\mathcal{S}_1 \subseteq \mathcal{S}_2 \qquad \text{or} \qquad \mathcal{S}_2 \subseteq \mathcal{S}_1.$$

Brief answer. If one is contained in the other, their union is the larger σ-algebra. Conversely, suppose neither contains the other. Choose $A \in \mathcal{S}_1 \setminus \mathcal{S}_2$ and $B \in \mathcal{S}_2 \setminus \mathcal{S}_1$. If $\mathcal{S}_1 \cup \mathcal{S}_2$ were a σ-algebra, then $A \cup B$ would belong to one of them. If $A \cup B \in \mathcal{S}_1$, then $B = (A \cup B) \setminus A \in \mathcal{S}_1$, a contradiction; the other case is symmetric.